

The 2025 GCPC Collegiate Programming Contest

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difficulty: medium

Time limit per test: 1s

Problem Alien Energy Balancing

On the distant planet Zorath, the energy grid is powered by an array of n energy cells, each with an initial energy level represented by an array e . The Zorathians have developed a new energy modulation device, but it only supports two types of energy adjustments, controlled by two fixed values x and y .

In one operation, the Zorathians can:

- Select a cell i (where $1 \leq i \leq n$) and increase its energy level e_i by x .
- Select a cell i (where $1 \leq i \leq n$) and increase its energy level e_i by y .

Note that x and y are constants and may be equal.

The stability of the energy grid is determined by the **range** of the energy levels, defined as $\max(e_i) - \min(e_i)$. For example:

- The range of the array $[1, 2, 3, 4]$ is $4 - 1 = 3$.
- The range of the array $[1, 1, 1]$ is $1 - 1 = 0$.

After performing any number of operations (possibly zero), the Zorathians want to minimize the range of the energy levels in the array. Your task is to compute the smallest possible range achievable.

Input:

- The first line contains a single integer t . The number of test cases.
- The second line contains three integers n , x , and y ($1 \leq n \leq 10^9$, $1 \leq x, y \leq 10^9$).

- The third line contains **n** integers **e₁**, **e₂**, ..., **e_n** ($1 \leq e_i \leq 10^9$), representing the initial energy levels.

Output:

- Output a single integer: the minimum possible range of the energy levels after any number of operations.

Example

Input:	Output:
2 1 1 1 1 4 7 7 1 1 2 6	0 3